SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA0301
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

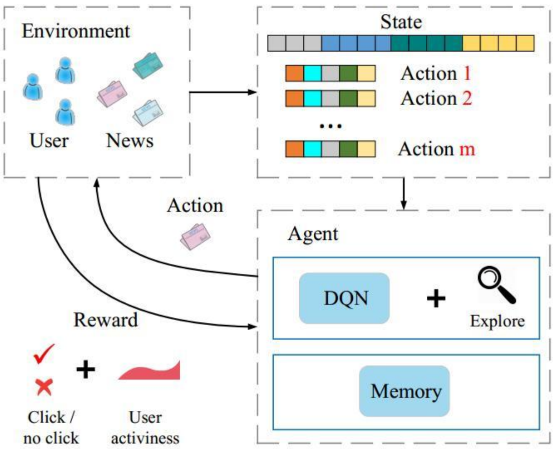
Case Study **Duration: 2hrs**

1. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. **(5 Marks)**

Introduction

A streaming platform (such as Netflix, Amazon Prime Video, or Disney+) aims to recommend movies that each user is most likely to enjoy. Since user preferences change over time, traditional recommendation systems may not always provide the best suggestions. Reinforcement Learning (RL) helps overcome this limitation by allowing the recommendation agent to learn continuously from user interactions and improve recommendations over time.

In RL, the recommendation system acts as an **agent** that interacts with the **environment** (users). The agent observes the user's current state, recommends a movie (action), receives feedback (reward), and updates its recommendation policy to maximize long-term user engagement.



RL Components

1. Agent

The recommendation engine that selects movies for users.

Responsibilities:

- Observe user behavior.
- Recommend suitable movies.
- Learn from user feedback.
- Improve future recommendations.

2. Environment

The environment includes:

- Users
- Movie database
- User interactions
- Viewing sessions

The environment provides feedback after every recommendation.

3. State Representation (S)

A state represents the current situation of a user.

The recommendation agent uses several features to represent the state.

User Profile

- Age
- Gender
- Location
- Preferred language

Viewing History

- Previously watched movies
- Genres watched frequently
- Number of movies watched
- Recently watched movies

User Preferences

- Favorite actors
- Favorite directors
- Favorite genres
- Favorite language

User Behavior

- Watch duration
- Completion percentage
- Watch frequency
- Search history

Device Information

- Mobile
- TV
- Laptop
- Tablet

Time Context

- Morning
- Afternoon
- Evening
- Weekend

Example State

State =
{

Age = 22
Preferred Genre = Action
Language = English
Recently Watched = Avengers
Completion Rate = 95%
Rating History = 4.8
Time = Evening
}

4. Action Space (A)

An action is the recommendation made by the system.

Possible actions include:

- Recommend Movie A
- Recommend Movie B
- Recommend Trending Movies
- Recommend Similar Movies
- Recommend New Genre
- Recommend Top-rated Movies
- Recommend Personalized Collection

Example

Current user likes Action movies.

Possible actions:

- Avengers Endgame
- Mission Impossible
- John Wick
- Mad Max
- Inception

Each recommendation is one action.

5. Transition Function

After recommending a movie, the user reacts.

Example:

Current State

User likes Action movies

↓

Action

Recommend John Wick

↓

User watches completely

↓

Next State

User preference for Action increases

The state changes based on user behavior.

6. Reward Function (R)

The reward function tells the agent whether its recommendation was good or bad.

A well-designed reward function is the key to successful RL.

Positive Rewards

User Activity	Reward
Watches entire movie	+10
Gives 5-star rating	+8
Likes movie	+7
Watches more than 80%	+6
Adds movie to watchlist	+4
Shares movie	+5
Recommends to friends	+6

Negative Rewards

User Activity	Reward
Skips recommendation	-2
Stops after 5 minutes	-5
Gives poor rating	-6
Removes from watchlist	-4
Closes application	-8

7. Policy

A policy determines how the agent selects movies.

Examples:

- Recommend highest predicted movie.
- Recommend similar movies.
- Recommend trending movies.
- Recommend unseen movies.

The policy keeps improving after every interaction.

8. Value Function

The value function estimates how beneficial a state is.

Example

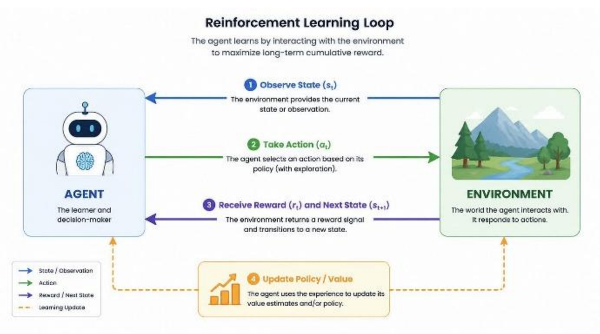
A user who consistently watches Action movies completely has a high-value state because recommending similar movies is likely to result in high rewards.

9. Objective of RL

The objective is to maximize cumulative rewards.

The system tries to maximize:

- Watch time
- User satisfaction
- Click-through rate
- Subscription renewal
- Daily active users
- Monthly active users



Analysis of Different Reward Signals

Different rewards influence the recommendation strategy in different ways.

1. Watch Completion Reward

Reward:

Complete movie = +10

Effect

- Recommends engaging movies.
- Increases average watch time.
- Improves session duration.

2. Rating-Based Reward

Reward

5 Stars = +8

Effect

- Learns personal preferences.
- Improves recommendation accuracy.
- Builds personalized profiles.

3. Click Reward

Reward

Movie clicked = +2

Effect

- Measures user interest.
- Encourages attractive recommendations.
- May not always indicate satisfaction.

4. Watchlist Reward

Reward

Added to Watchlist = +4

Effect

- Indicates future interest.
- Helps recommend sequels or related content.

5. Skip Penalty

Reward

Skipped = -2

Effect

- Avoids recommending irrelevant movies.
- Reduces recommendation errors.

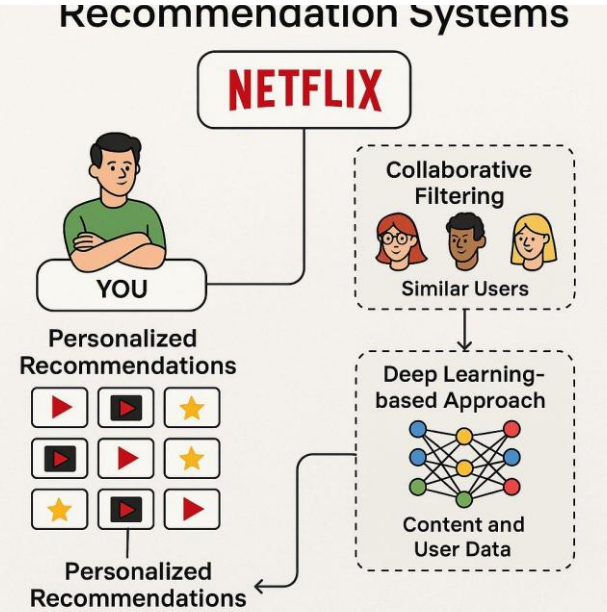
6. Early Exit Penalty

Reward

Stopped after few minutes = -5

Effect

- Strong indication of poor recommendation.
- Encourages better movie matching.



Impact of Reward Signals on User Engagement

Reward Signal	**Impact**
Watch completion	Increases viewing time
Ratings	Improves recommendation quality
Likes	Learns user interests quickly
Watchlist	Encourages future viewing
Shares	Expands content popularity
Skip penalty	Removes poor recommendations
Early exit penalty	Prevents repeated mistakes

Exploration vs Exploitation

One of the biggest challenges in Reinforcement Learning is balancing exploration and exploitation.

Exploration

The agent recommends unfamiliar movies.

Example:

User watches only Action movies.

The system recommends

- Drama
- Mystery
- Documentary

Advantages

- Discovers hidden interests.
- Increases recommendation diversity.
- Prevents boredom.
- Learns new preferences.

Disadvantages

- User may dislike recommendations.
- Lower short-term rewards.

Exploitation

The agent recommends movies already known to match the user's interests.

Example

User likes Action.

Recommend:

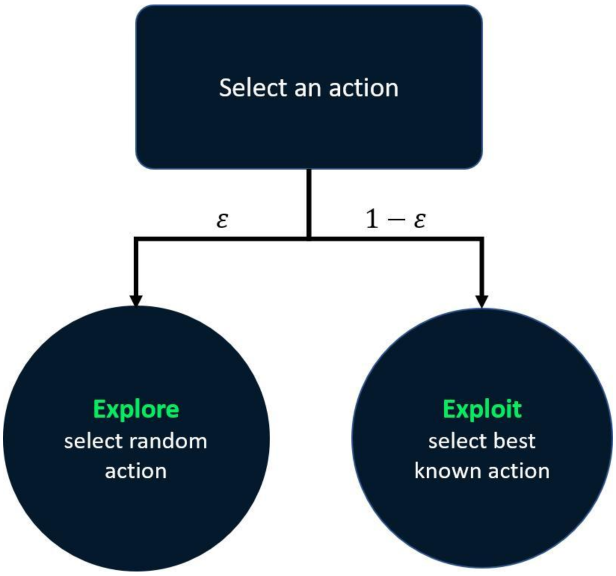
- Avengers
- John Wick
- Extraction
- Mission Impossible

Advantages

- Higher immediate reward.
- Better click rate.
- Better watch completion.
- Higher satisfaction.

Disadvantages

- Recommendations become repetitive.
- User misses discovering new content.
- Creates a "filter bubble."



Trade-off Between Exploration and Exploitation

Exploration	**Exploitation**
Finds new interests	Uses known preferences
Long-term learning	Short-term satisfaction
Higher diversity	Higher accuracy
Temporary lower rewards	Higher immediate rewards
Prevents repetitive recommendations	May reduce diversity

Balancing Both Approaches

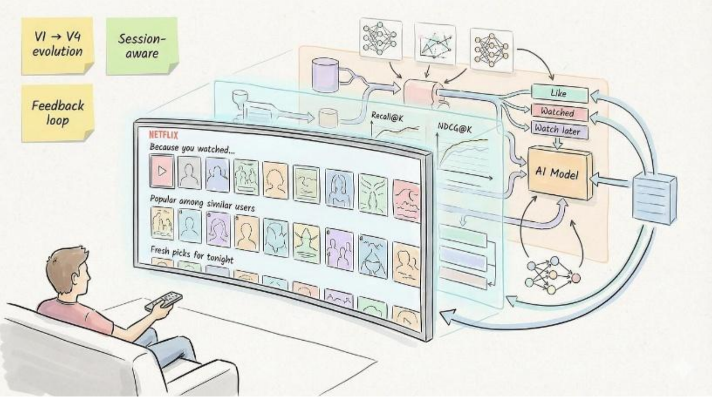
A common strategy is the **ε-greedy (epsilon-greedy) policy**:

- **90% of the time:** Recommend movies the user is most likely to enjoy (exploitation).
- **10% of the time:** Recommend new or less familiar movies (exploration).

This balance helps the platform maintain high user satisfaction while continuously learning evolving preferences.

Advantages of RL-Based Movie Recommendation

- Learns continuously from user feedback.
- Adapts to changing preferences over time.
- Provides highly personalized recommendations.
- Increases watch time and user engagement.
- Improves click-through rate and subscription retention.
- Handles dynamic user behavior better than static recommendation methods.



Limitations

- Requires large amounts of interaction data.
- Training can be computationally expensive.
- Poorly designed reward functions can lead to suboptimal recommendations.
- Exploration may temporarily reduce user satisfaction.

- Cold-start users (with little viewing history) remain challenging.

Conclusion

Reinforcement Learning enables a streaming platform to continuously improve its movie recommendations by learning from user interactions. By representing user information as **states**, selecting movie recommendations as **actions**, and evaluating outcomes through carefully designed **reward functions**, the system maximizes long-term user engagement rather than only immediate clicks. A balanced exploration–exploitation strategy ensures that users receive both familiar and new content, resulting in higher satisfaction, increased watch time, better retention, and a more personalized streaming experience.

Code of Conduct:

I Malli Chandana(19242250)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Malli Chandana

RUBRICS FOR CASESTUDY

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understandi ng with most issues identified	Basic understanding ; some issues missed	Poor understand ing; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured , and easy to understand	Organized and understanda ble presentatio n	Limited clarity and structure	Poor clarity; difficult to understand